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Potential-dependent superlubricity of stainless steel and Au(1 1 1) using a water-in-surface-active ionic liquid mixture

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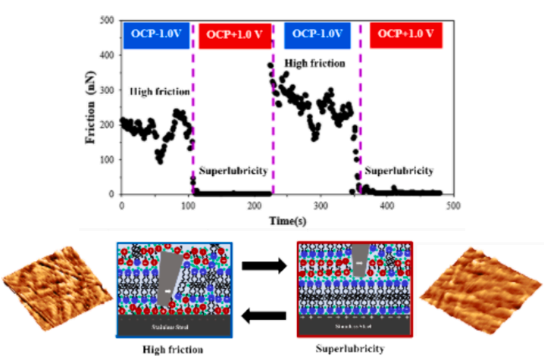
GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: The friction and interfacial nanostructure of a water-in-surface-active ionic liquid mixture, 1.6 M 1-butyl-3-methylimidazolium 1,4-bis-2-ethylhexylsulfosuccinate ([BMIm][AOT]), can be tuned by applying potential on Au(1 1 1) and stainless steel.

Experimental: Atomic force microscopy (AFM) was used to examine the friction and interfacial nanostructure of 1.6 M [BMIm][AOT] on Au(1 1 1) and stainless steel at different potentials.

Findings: Superlubricity (vanishing friction) is observed for both surfaces at OCP+1.0 V up to a surface-dependent critical normal force due to $[AOT]^-$ bilayers adsorbing strongly to the positively charged surface thus allowing AFM tip to slide over solution-facing hydrated anion charged groups. High-resolution AFM imaging reveals ripple-like features within near-surface layers, with the smallest amplitudes at OCP+1 V, indicating the highest structural stability and resistance to thermal fluctuations due to highly ordered boundary $[AOT]^-$ bilayers

Abbreviations: [BMIm][AOT], 1-butyl-3-methylimidazolium 1,4-bis(2-ethylhexoxy)-1,4-dioxobutane-2-sulfonate; AFM, atomic force microscopy; ILs, ionic liquids; HOPG, highly oriented pyrolytic graphite; SAIL, surface-active ionic liquid; ECW, electrochemical window; OCP, open circuit potential; *B*, modulus of layer compression; *F*, normal force; *l*, separation; F_n, normal force at the nth minima; l_n, separation at the nth minima; *R*, contact radius.

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templating robust near-surface layers. Exceeding the critical normal force at OCP+1.0 V causes the AFM tip to penetrate the hydrated $[AOT]^-$ layer and slide over alkyl chains, increasing friction. At OCP and OCP-1.0 V, higher friction correlates with more pronounced ripples, attributed to the rougher templating $[BMIm]^+$ boundary layer. Kinetic experiments show that switching from OCP-1.0 V to OCP+1.0 V achieves superlubricity within 15 s, enabling real-time friction control.

1. Introduction

Friction is the force that resists relative motion between sliding surfaces. Around 40 % of the energy dissipated in mining operations, and 33 % of fuel consumed in vehicles, are lost to friction and associated wear [1,2]. The economic cost of this is enormous, and as a consequence, considerable research effort is dedicated to the development of new and advanced high-performance lubricants to reduce friction.

Superlubricity refers to situations where friction coefficients (the ratio of the friction force to the normal force) are below 0.01, meaning friction is nearly eliminated [3–6]. Superlubricity has been widely reported for solid lubricants with 2D lamellar structures such as graphite [5,7–10], h-BN [11,12], and MoS_2 [13], due to their atomically smooth surfaces, weak interlayer interactions, and lattice incommensurability between sliding surfaces. However, for solid lubricants, there is a significant "wearing in" period during which the material adsorbs to the sliding surfaces before friction decreases [8].

Superlubricity has also been found in surfactant, lipid and polymer aqueous systems [14–18], especially those with lamellar phases [19–22]. These lamellar phases comprise stacks of 2D amphiphile bilayers separated by solvent, allowing low-resistance sliding similar to solid 2D materials [19–22]. However, the exact lubrication mechanisms remain unclear, with debates on whether slip occurs within bilayers or solvent layers [22–24]. Recently, hydrated ionic surfactant headgroups were shown to be highly lubricating, underpinned by the fluidity of the water molecules in its hydration sheath, which facilitate the significant reduction in friction [14,25–28]. Rupturing of the bilayer structure on the surface increases friction owing to the loss of the hydration lubrication mechanism mediated by the headgroups on the outer half of the bilayers [28–31].

Ionic liquids (ILs), pure salts with low melting points and designable nature, are promising lubricants [32–35] due to their non-volatility, non-flammability, high thermal stability [36–38] and, for electrotunable lubrication [39], a wide electrochemical window. Many ILs have at least one amphiphilic ion – often the cation – leading to well-defined sponge-like bulk nanostructures [40]. Near a surface, this bulk nanostructure becomes flatter and more ordered perpendicular to the surface, analogous to surface-induced sponge to lamellar phase transitions in aqueous surfactant systems [41].

Surface force apparatus and atomic force microscopy (AFM) have been used to probe near-surface IL structure [42–52], revealing that weak lateral cohesive forces allow displacement of near-surface structures at low normal forces [53]. High resolution video-rate AFM shows that for nanostructured ILs [54–56] the near-surface structures are flattened IL aggregates [57,58]. The boundary layer ions bind to the surface strongly through a variety of interactions, resisting "squeezing out" up to high normal forces [59]. This strong adsorption makes ILs promising boundary lubricants [51,60], with friction predominantly determined by the cation-to-anion ratio and ion structures, which can be tuned by surface potential [49,61–67]. Negative polarization has achieved superlubricity on HOPG and Au(1 1 1) due to the formation of a highly ordered boundary layer enriched in cations with relatively long alkyl chains [49,64,66]. However, to date, electrotunable IL superlubricity has not been demonstrated at a positively charged surface or for industrially applicable surfaces like stainless steel.

Surface-active ionic liquids (SAILs), a subclass of ILs containing surfactant-like ions, combine the useful properties of both ILs and surfactants and have been investigated for diverse applications ranging from electrochemistry to drug delivery [52,68–71]. Recently, we have studied the bulk phase sequences of one SAIL, 1-butyl-3-methylimidazolium 1,4-bis(2-ethylhexoxy)-1,4-dioxobutane-2-sulfonate ([BMIm][AOT]) upon dilution with water [72]. No free water has been detected up to a water concentration of 25 vol% (1.4 M [BMIm][AOT]), indicating these SAIL-water mixtures may behave like water-in-salt electrolytes, which are highly concentrated aqueous electrolytes with salt-to-water molar ratios greater than one [73,74]. The well-defined lamellar structures in these mixtures make them promising candidates for lubrication [72]. In this study, AFM friction and force curve measurements are used to investigate the lubrication performance and interfacial nanostructure of one water-in-SAIL mixture with a bulk lamellar structure (1.6 M [BMIm][AOT]), on Au(1 1 1) and stainless steel as a function of potential.

2. Material and methods

1-butyl-3-methylimidazolium chloride ([BMIm]Cl, purity > 99 %) was purchased from IoliTec (Heibronn, Germany) and used without further purification. Sodium 1,4-bis(2-ethylhexoxy)-1,4-dioxobutane-2-sulfonate (Na[AOT], purity > 99 %), dichloromethane (purity > 99 %), and silver nitrate (purity > 99 %) were all purchased from Sigma-Aldrich and used as received. Au(1 1 1) surfaces (a 300 nm thick atomically smooth gold film on mica) were purchased from Georg Albert PVD – Beschichtungen. [BMIm][AOT] was synthesized using a previously described method.[75,76] Typically, [BMIm]Cl and Na[AOT] were mixed in dichloromethane. The NaCl by-product was precipitated by the addition of silver nitrate, then filtered out and rinsed with water. Upon evaporation of the dichloromethane solvent, the viscous [BMIm][AOT] was produced.

Cyclic voltammetry (CV) was recorded at a scan rate of 100 $mV{\cdot}s^{-1}$ using a PGSTAT101 potentiostat (Metrohm Autolab, Gladstone, NSW, Australia) controlled by a software (Nova 1.11) at 294 ± 1 K. The working electrode used in this study was a micro disk (micrometre sized) platinum electrode, a silver wire was used as both the counter and the reference electrodes. The sample was dried under high vacuum for 30–60 mins before tests in order to remove water and other residue solvents, which affect the CV results significantly.[77,78] The electrochemical cell was housed inside an aluminium Faraday cage to reduce electromagnetic interferences in all experiments. The stability of CV signals was confirmed through multiple consecutive scans with identical CV shapes.

A Veeco Nanoscope IV AFM was used for both friction and normal force measurements. Au(1 1 1) surfaces and sharp silicon AFM tips (NSC36, Mikromasch) with spring constants of 0.7 ± 0.3 $N{\cdot}m^{-1}$ were cleaned with ethanol and MilliQ water, followed by UV-Ozone for 10 min before use. The Bruker AFM fluid cell was cleaned with ethanol, rinsed with MilliQ water, and dried under a nitrogen stream before each experiment. Surface potentials were adjusted using a WaveNano® potentiostat.

Friction forces were obtained by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled. The scan size was 500 nm, and scan rate was 6 Hz. The lateral deflection signal (i.e. cantilever twist) was converted to friction using a customized function produced in MATLAB which takes into account the torsional spring constant and the geometrical dimensions of the cantilever [79], as detailed in the SI. Normal force vs separation curves were collected by moving the surface towards the

AFM tip and detecting the cantilever deflection as a function of separation. The ramp size and rate were 50 nm and 0.2 Hz, respectively. Standard methods were used to convert deflection vs separation data to normal force vs separation curves [80]. Each experiment was repeated at least three times on different positions of the surface.

The AFM images were captured using an Asylum Research Cypher VRS Atomic Force Microscope (Oxford Instrument) using tapping mode, where the cantilevers were oscillated at (or close to) their resonant frequency with a blueDrive laser. Stiff cantilevers (FS1500AuD, Asylum Research) with a relatively high nominal spring constant of 6.0 ± 0.5 N/m were utilised to image the IL adsorbed layers on both Au (1 1 1) and stainless steel surfaces.

3. Results and discussion

The water-in-[BMIm][AOT] mixture (1.6 M [BMIm][AOT]) has an electrochemical window (ECW) spanning from −1.75 V to +1.75 V (**Fig. S1** in the supporting information), which is significantly wider than that of water (1.23 V) [81]. The enhanced ECW confirms that this [BMIm][AOT]-water mixture is a water-in-salt electrolyte in which water molecules are bound to cations and anions which impede the decomposition reactions of bulk water [73,74]. This wide ECW makes it feasible to evaluate the potential-dependent lubricity.

3.1. Friction on Au(1 1 1)

The nanoscale friction of 1.6 M [BMIm][AOT] on Au(1 1 1) as a function of potential measured using AFM is shown in Fig. 1. The friction coefficients are listed in Table 1. Repeated friction curves are presented in **Fig. S2**, which show the errors are smaller than the observed variations in friction trends. At the open circuit potential (OCP), when the normal force increases from 0 nN to 80 nN, the friction increases linearly from ~0 nN to ~40 nN (Fig. 1**a**), which is ~60 % lower than that observed in air and 50 % lower than in water. (**Fig. S3**). Since Au(1 1 1) is marginally negatively charged at the OCP, the boundary layer, although composed of mixed cations and anions, is slightly enriched in $[BMIm]^+$ cations with the butyl chains pointing towards the bulk. The reduction in friction indicates that this $[BMIm]^+$ enriched boundary layer helps to resist the direct contact between surfaces and lubricates the surface.

When Au(1 1 1) is negatively charged (OCP−0.5 V and OCP−1.0 V), the friction for 1.6 M [BMIm][AOT] is lower than that at OCP, especially when the normal force is higher than 30 nN, as shown in Fig. 1**a**. This reduction is attributed to a better-defined and more strongly bound layer of $[BMIm]^+$ due to stronger electrostatic interactions between the $[BMIm]^+$ and the negatively charged Au(1 1 1) surface. When the potential is decreased from OCP−0.5 V and OCP−1.0 V, the friction is essentially unchanged. This suggests that either the boundary layer is saturated in $[BMIm]^+$ at OCP−0.5 V, or additional $[BMIm]^+$ incorporated into the boundary layer does not make it measurably more robust.

Strikingly, at both positive potentials investigated (OCP+0.5 V and OCP+1.0 V), superlubricity is observed in the low normal force regime for 1.6 M [BMIm][AOT], *c.f.* Fig. 1**b** and Table 1**.** When the normal force increases above a threshold, the friction jumps abruptly to a high level and increases further with the normal force. Similar friction discontinuities have been found in other aqueous and IL surfactant systems, which are mainly due to the rupture of the boundary layer.[16,51,59,82,83] The magnitude of the critical normal force increases with the surface potential, from 11 nN at +0.5 V to 36 nN at +1.0 V. This is because the boundary $[AOT]^-$ bilayer is more robust at a more positive potential due to stronger electrostatic interactions of $[AOT]^-$ headgroups with the surface, and thus can resist rupture up to a higher normal force. The sudden jump in friction is also higher at +1.0 V than at +0.5 V, indicating stronger lateral interactions boundary layer $[AOT]^-$. Consequently, more energy is required to brush aside the boundary layer when the surface is more positively charged. However, at both positive potentials, after the $[AOT]^-$ bilayer is ruptured, the friction coefficients are comparable and still substantially lower than that in air, *cf.* Table 1, which implies that a layer remains between the Au(1 1 1) surface and the AFM tip.

To the best of our knowledge, this is the first time superlubricity has been observed at a positively charged metal surface at the nanoscale. Previous studies of pure [BMIm][AOT] on HOPG have shown that the boundary layer is enriched in $[AOT]^-$ at positive potentials.[52] Superlubricity for 1.6 M [BMIm][AOT] at positively charged Au(1 1 1) is attributed to the formation of well-defined boundary $[AOT]^-$ bilayers. Hydrating water molecules may also facilitate superlubricity, as shown in previous studies for surfactant and polymer aqueous solutions, due to the fluidity of water within the hydration layers even under strong confinement.[25,26,28].

3.2. Lateral and normal structure on Au(1 1 1)

To further understand the superlubricity induced by the $[AOT]^-$ bilayers and identify the layer structure above the critical normal force at positive potentials, lateral interfacial structure and normal force curves of 1.6 M [BMIm][AOT] on Au(1 1 1) as a function of potential have been investigated by AFM, *c.f.* Fig. 2 and Fig. 3.

Fig. 2 shows that the structure of the bare Au(1 1 1) surface (Fig. 2**b** inset and **Fig. S4**) is replaced by ripple-like interfacial structures at all three potentials. Due to the limitations of current-generation AFM, the absolute separation between surfaces could not be determined, making it uncertain which layer was imaged. However, the observed ripple-like structures differ from typical IL boundary (Stern) layers enriched in individual ions or ion bilayers[64,84] but resemble previously observed lamellar phases [85–87]. This suggests that the AFM images depict one of the near-surface lamellar layers rather than the boundary ion layer or

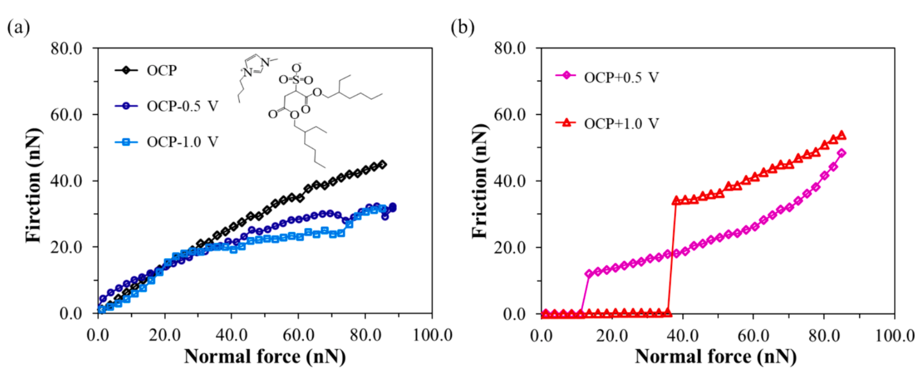

Fig. 1. Friction vs normal force of 1.6 M [BMIm][AOT] on Au(1 1 1) at (a) OCP, OCP−0.5 V and OCP−1.0 V and (b) OCP+0.5 V and OCP+1.0 V, respectively. The molecular structure of [BMIm][AOT] is shown as an inset in (a). Errors fall within ±10 %.

Table 1
Friction coefficient of 1.6 M [BMIm][AOT] on Au(1 1 1) at different potentials.

	Air	OCP	OCP−0.5 V	OCP−1.0 V	OCP+0.5 V		OCP+1.0 V	
					μ_1	μ_2	μ_1	μ_2
μ	**1.125**	**0.524**	**0.312**	**0.312**	**0.001**	**0.442**	**0.001**	**0.431**

* μ_1 and μ_2 are friction coefficients below and above the critical normal force shown in Fig. 1**b**. Errors fall within ±10 %.

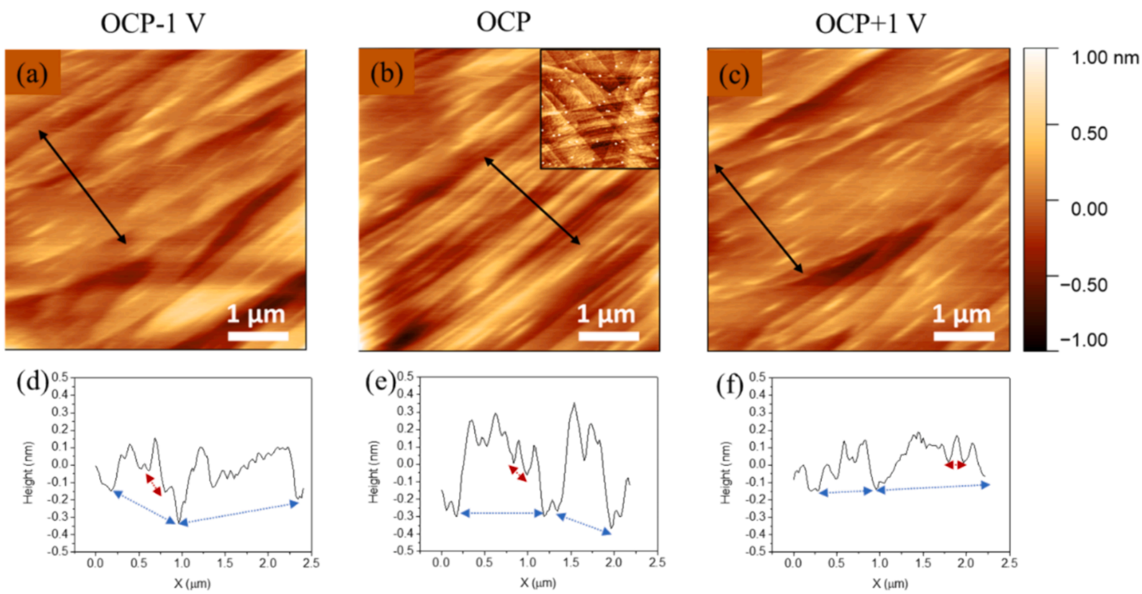

Fig. 2. Topographic AFM images and height profiles of 1.6 M [BMIm][AOT] on Au(1 1 1) at (a, d) OCP−1.0 V, (b, e) OCP, and (c,f) OCP+1.0 V, respectively. The inset in (b) shows the topographic image of bare Au(1 1 1) in air at the same scale. The black arrows in (a)-(c) indicate the positions to extract height profiles shown in (d)-(f). The blue arrows in (d)-(f) indicate valley to valley distances of large ripples, and the red arrows indicate the valley-to-valley distances of small ripples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

bilayer at Au(1 1 1). Estimating the normal force applied on the AFM tip during tapping mode imaging is challenging due to the difficulty in achieving the quality factor (Q) in a viscous liquid [88], such as the 1.6 M [BMIm][AOT] here, particularly when the AFM cantilever is tuned by a blue drive laser instead of a piezo. However, it is reasonable to assume that the applied normal force on AFM tip during imaging is lower than the critical normal force found at positive potentials in Fig. 1. Given the consistency of the normal force data presented later, it is likely that the layer the AFM tip slides on in friction experiments has similar structure to the one shown in Fig. 2.

Large ripples overlaid with much smaller and narrower ripples are visible at all three potentials (Fig. 2**a-c**). Even the small ripples have valley-to-valley distances two orders of magnitude larger than the molecular dimensions of ions and water present in 1.6 M [BMIm][AOT], and are suggestive of thermal fluctuations of lamellae in the bulk. The small ripples are most prominent at OCP, and diminish in amplitude at OCP−1.0 V and OCP+1.0 V. The large ripples may also result from fluctuations or may indicate the presence of edge defects in the lamellar phase [89]. They also decrease in amplitude as the potential changes from OCP to OCP−1.0 V and OCP+1.0 V, which is accompanied by an increase in their valley-to-valley distance as shown in the height profiles (Fig. 2**d-f**).

Combining the observations from Fig. 1 and Fig. 2 reveals that the reduction trends in friction and ripple-like structures are consistent: both are highest at OCP, lower at OCP−1.0 V, and lowest at OCP+1.0 V. The presence and behaviour of these ripples indicate the structural stability of the near-surface lamellar layers and their resistance to thermal fluctuations, influencing friction and energy dissipation. At OCP, the more pronounced ripples suggest weaker resistance to thermal fluctuations due to a less-defined near-surface lamellar layer templated from a less ordered boundary layer. This allows thermal fluctuations to disrupt the layers more easily by introducing temporary distortions or defects, resulting in increased friction and energy dissipation during sliding. In contrast, less prominent ripples, especially the small ones at biased surfaces, indicate the formation of more robust boundary layers due to stronger electrostatic interactions. This produces well-defined near-surface lamellar layers with improved resistance to thermal fluctuations and deformation during sliding, reducing friction and energy dissipation. Superlubricity is observed at OCP+1 V because the [AOT]$^-$ bilayer structures in the boundary region are highly ordered, which templates smooth, robust near-surface lamellar layers with the weak thermal fluctuations and lowest energy dissipation.

Normal force curves for 1.6 M [BMIm][AOT] on Au(1 1 1) as a function of potential are shown in Fig. 3. A single representative normal force curve is shown instead of average heat maps because these can obscure important features at small normal forces. The term "apparent separation" is used on the *x* axis because at a distance of zero, the AFM tip may press against either the surface or ions that cannot be displaced [90]. Previous AFM normal force [64], friction [91], and AFM imaging data [57,84,92,93] have shown that the boundary layer IL ions remain in place up to high normal forces.

At a wide separation, the AFM tip detects zero normal force as it is not sensitive to the bulk structure. As the AFM tip approaches the surface, it measures a sequence of steps (discontinuities) at regular intervals. The maximum height of each step represents the force necessary to break through a layer, referred to as the "rupture force" or "push-through force". More structured layers usually require higher push-through forces. As the apparent separation diminishes, the push-through force increases since the liquid interfacial nanostructure is better defined near the surface. The step width is typically affected by

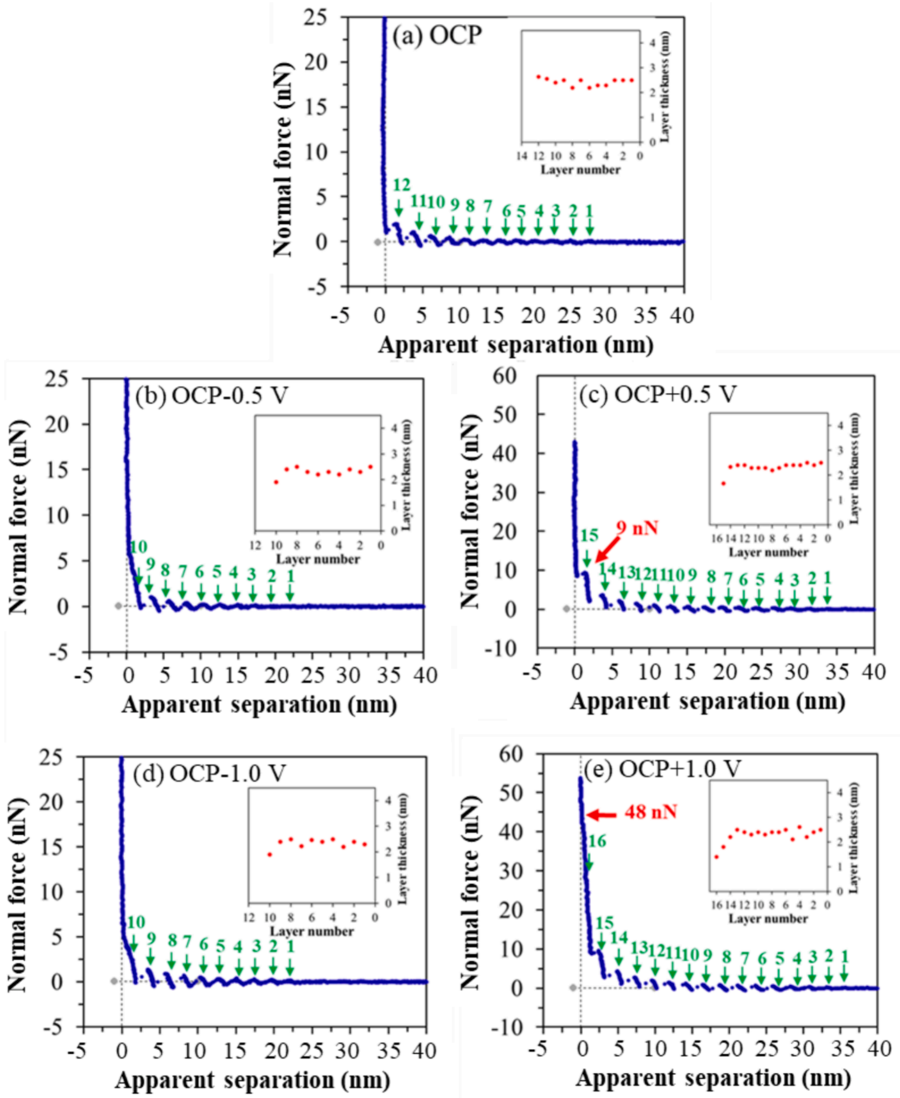

Fig. 3. Representative normal force vs apparent separation profiles at Au(1 1 1) surface for 1.6 M [BMIm][AOT] at (a) OCP, (b) OCP−0.5 V, (c) OCP+0.5 V, (d) OCP−1.0 V, and (e) OCP+1.0 V, respectively. The green arrows highlight the detectable layers from the bulk to the surface. Insets display the thickness of each detectable layer. The red arrows in (c) and (e) mark the push-through forces of the innermost layer. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the dimensions of the cation, anion or ion pairs displaced when a layer is pushed through, thus providing insights into the layer composition [94]. The actual interfacial structure may be more complex than the simplified flat layers suggested by AFM force measurements, potentially involving multilayer structures like micelles or nano-onions. However, AFM force curves represent an average interaction over the tip contact area (hundreds of nm^2), consolidating the force profiles of intricate structures into data resembling simplified flat layers, even though the actual interfacial nanostructure may be more sophisticated.

The near-surface nanostructure of 1.6 M [BMIm][AOT] exhibits notably more layers than pure [BMIm][AOT] [52] and other pure SAIL systems [75], where less than four layers are typically detected. Pure [BMIm][AOT] forms a lamella-like structure at the surface which decays to the 3D sponge-like structure within less than 10 nm [52]. By contrast, 1.6 M [BMIm][AOT] forms a lamellar phase both in the bulk and at the surface, leading to a slower rate of structural decay and hence more layers can be detected by AFM. The large number of steps detected is similar to prior studies of conventional surfactant lamellar phases [87]. When the tip surface distance is ~30 nm in this system (Fig. 3), the lamellar stacks are no longer influenced by the surface and zero force is measured.

Fig. 3**a** shows that at OCP, as the AFM tip approaches Au(1 1 1), it encounters a weak layer at ~27.5 nm, which is pushed through at a force of ~0.2 nN. The AFM tip jumps inward and encounters another layer at ~25 nm. This process continues 12 times until the AFM tip ruptures the innermost discernible layer at a force of 2.5 nN. The widths of all detected layers at OCP (~2.5 nm, Fig. 3**a** inset) are consistent with the repeat distance of the lamellar phase in the bulk (2.6 nm), akin to previous findings in surfactant aqueous systems [87,95,96], and indicates the $[AOT]^-$ bilayer structure is retained at the interface [72].

At negative potentials (OCP−0.5 V and OCP−1.0 V), ~10 layers are detected, *c.f.* Fig. 3**b** and **d**. The thicknesses of the layers are comparable to those at OCP. The forces to push through the final layer are ~5 nN which is twice as high as OCP. These results indicate that compared to OCP, the boundary layers enriched in $[BMIm]^+$ cations at negative potentials are more ordered and better template subsequent lamellae in the near-surface region, which require higher forces to push through. This is in accordance with the friction data (Fig. 1) and AFM images (Fig. 2). Typically, charged surfaces tend to exhibit more near-surface layers compared to neutral surfaces for pure SAILs and other ILs [64,97,98].

However, in this system, slightly fewer layers are detected at negative potentials than at OCP. This suggests that higher concentrations of $[AOT]^-$ anions in the boundary layer may be more effective at facilitating the formation of near-surface layers compared to $[BMIm]^+$ cations, aligning with the detection of more near-surface layers at positive potentials where the boundary layer is more enriched in $[AOT]^-$ anions.

At positive potentials (OCP+0.5 V and OCP+1.0 V), more layers (>15 layers) are detected. In this case, the boundary is likely enriched in $[AOT]^-$ anions with the charged headgroups attracted to the surface and alkyl chains towards the bulk. Due to the strong solvophobic interactions of long alkyl chains, an additional layer of mainly $[AOT]^-$ anions forms on the adsorbed layer with headgroups facing away from the surface. Thus, a well-defined boundary $[AOT]^-$ bilayer forms at the positively charged surface, *c.f.* Fig. 4**a**, which templates more robust subsequent layers, consistent with the weaker ripple-like structures shown in Fig. 2 at this potential.

The final push-through forces observed at the two positive potentials in Fig. 3**c** and **e** are generally consistent with the critical normal forces identified in the friction measurements, beyond which the friction increases abruptly (Fig. 1). The thickness of the final detectable layer is ~1.2 nm, which is about half of that of the outer layers, as seen in the insets of Fig. 3**c** and **e**. This is consistent with the superlubricity detected in the low normal force regime of Fig. 1**b** resulting from the AFM tip sliding over the hydrated headgroups of the $[AOT]^-$ bilayer, *c.f.* Fig. 4**a**. This aligns with aqueous surfactant systems, which has found that hydrated ions contribute to low friction due to the high fluidity of water molecules [14,27,28,99]. Once the normal force exceeds the threshold, at both positive potentials, friction coefficients increase significantly but are still lower than that in air, *c.f.* Table 1. Therefore, when the normal force is higher than the critical threshold, the AFM tip pushes through the upper half of the $[AOT]^-$ bilayer and slides over the alkyl chains of the lower half of the $[AOT]^-$ bilayer, *c.f.* Fig. 4**b**.

In contrast, at negative potentials, the absence of high push-through force and thinner innermost layer suggests that the $[BMIm]^+$ enriched boundary layer adheres strongly to the negatively charged surfaces, with the imidazolium head towards the surface and the butyl chain towards bulk, which cannot be squeezed out up to high force, *c.f.* Fig. 4**c**. No superlubricity is observed at negative potentials likely because the cohesive interactions between the short butyl chains are weaker compared to those between longer alkyl chains of $[AOT]^-$ at positive potentials, leading to a relatively poorly formed boundary layer and thus higher friction.

A weak attractive background force, likely due to increased ordering of the near-surface structured phase, is detected for the [BMIm][AOT]-water mixture on Au(1 1 1) (Fig. 3 and **Fig. S5**) at all potentials, similar to that found for conventional surfactant aqueous solutions [87,95,100].

For lamellar structures, the modulus of layer compression (B), which quantifies the resistance of the lamellar layers to compression, can be extracted directly from the normal force curves using [96]:

$$(F - F_n)\frac{l_n}{\pi R} = B(l_n - l)^2 \quad (1)$$

where F is the normal force, l is the separation, F_n and l_n are the normal force and separation at the n^{th} minima, and R is the contact radius. B has been found to be independent of n and can be extracted from the slope of the plot $(F\text{-}F_n)/R$ vs $(l_n\text{-}l)^2$, with negligible effect from the attractive background [96].

B values for 1.6 M [BMIm][AOT] on Au(1 1 1) at OCP, OCP-1 V and OCP+1 V were extracted from force curve data (Fig. 3) using **Eq. (1)**, *c.f.* Table 2. The $(F\text{-}F_n)/R$ vs $(l_n\text{-}l)^2$ plots (**Fig. S6**) show that data for different near-surface $[AOT]^-$ bilayers fall onto the same master curve at each potential, indicating B is independent of layer number, separation, and attractive background [96]. The fitted B values are lowest at OCP, higher at OCP-1 V and reach the maximum at OCP+1 V. This means the near-surface $[AOT]^-$ bilayers are most rigid (resistant to compression) at +1.0 V, consistent with superlubricity up to high normal force (Fig. 1**b**).

3.3. Friction and structure on stainless steel

The interfacial behaviour of 1.6 M [BMIm][AOT] was further evaluated on a commercially available stainless steel surface, *c.f.* Fig. 5, Table 3 **and Fig. S7**. Despite the higher surface roughness (**Fig. S4**), the trends for friction and interfacial nanostructures observed on stainless steel (Fig. 5) are similar to Au(1 1 1) (Fig. 3). This indicates that for 1.6 M [BMIm][AOT] the lamellar structure is so well defined that substrate roughness has little effect on nanostructure and lubricity. Consequently, superlubricity is also observed on stainless steel.

At OCP and OCP−1.0 V on stainless steel, the friction is close to zero below a normal force of 20 nN. This is attributed to the AFM tip sliding on hydrated near-surface $[AOT]^-$ bilayers. At higher normal forces these layers are penetrated and the tip contacts boundary layer $[BMIm]^+$ cations. However, the roughness of the stainless steel surface (~0.8 nm, Fig. S4b) [48] is larger than the molecular dimensions of $[BMIm]^+$ making AFM tip-surface contact likely, leading to higher friction. This is consistent with the AFM images in Fig. 5**c** and **d** which reveal surface

Table 2
B values for 1.6 M [BMIm][AOT] on Au(1 1 1) at different surface potentials.

Surface potential	OCP−1.0 V	OCP	OCP+1.0 V
B (nN/nm^2)	0.29	0.14	1.0

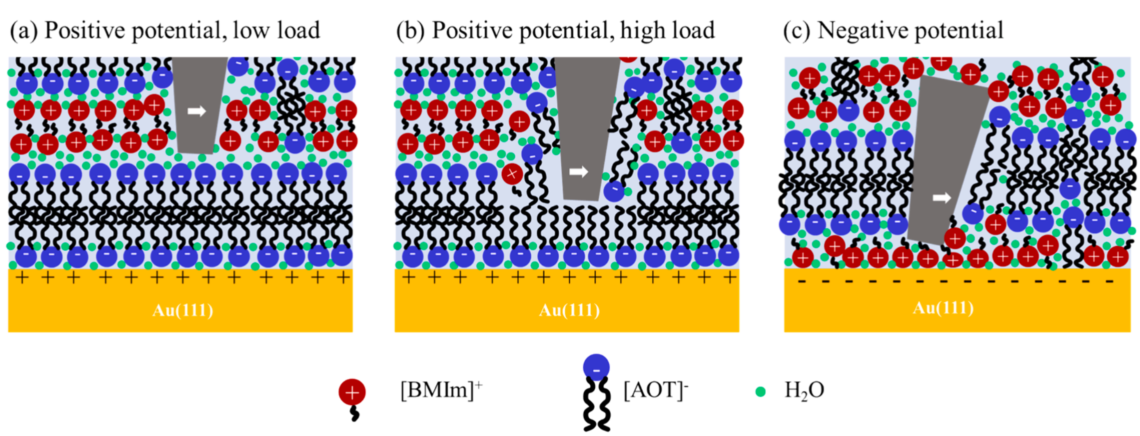

Fig. 4. Schematics of the AFM tip (in grey) sliding on the boundary layer enriched in (a) $[AOT]^-$ monolayer on a positively charged surface at a normal force lower than the critical normal force, (b) hydrated $[AOT]^-$ bilayer on a positively charged surface at a normal force higher than the critical normal force, and (c) hydrated $[BMIm]^+$ cations on a negatively charged surface.

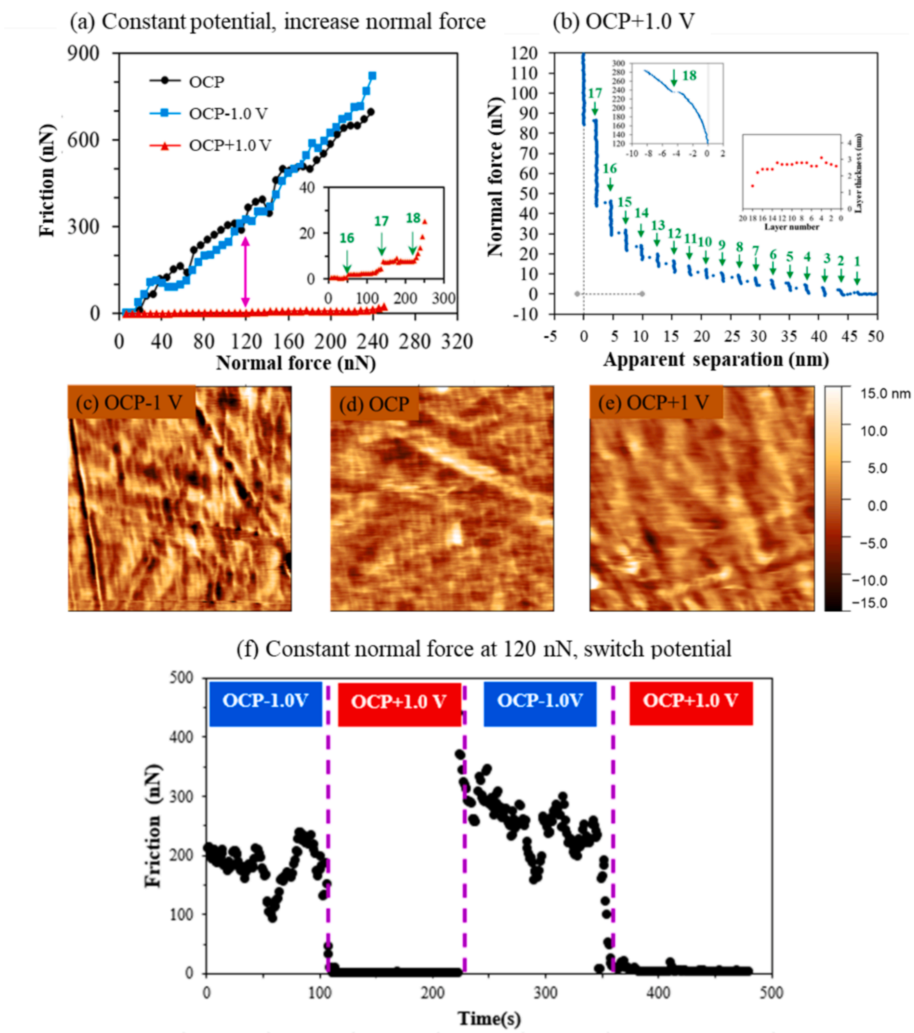

Fig. 5. (a) Friction vs normal force of 1.6 M [BMIm][AOT] on stainless steel at OCP, OCP−1.0 V, OCP+1.0 V, respectively. The inset provides a zoomed-in view of the friction data at OCP+1.0 V, highlighting the critical normal force of superlubricity. The magenta arrow marks the normal force of 120 nN, which is held consistent in (c). (b) Representative normal force vs apparent separation curves on stainless steel for 1.6 M [BMIm][AOT] at OCP+1.0 V. The green arrows highlight the detectable layers from the bulk to the surface. The force curve in the inset deviates from the vertical in the high force regime because the applied normal force surpasses the compliance of the cantilever. The thickness of each detectable layer is also shown in the inset. (c-e) Topographic AFM images of 1.6 M [BMIm][AOT] on stainless steel at: OCP−1.0 V, OCP, and OCP+1.0 V, respectively. (f) Variation in friction as the surface potential alternates between OCP−1.0 V and OCP+1.0 V at a constant normal force of 120 nN. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Friction coefficient of 1.6 M [BMIm][AOT] on stainless steel at different potentials.

	OCP	OCP−1.0 V	OCP+1.0 V		
			μ_1	μ_2	μ_3
μ	3.400	3.400	0.003	0.009	0.009

* The μ_1, μ_2 and μ_3 are the friction coefficients on each flat region in friction shown in Fig. 5**a**. Errors fall within ±10 %.

asperities and imperfections beneath the ripple-like structures of the lamellae**.**

At OCP+ 1.0 V, zero friction is detected up to a normal force of 50 nN on stainless steel (Fig. 5**a**). Then, the friction suddenly increases to 1.8 nN, remaining almost constant up to a normal force of 120 nN. Beyond that, the friction jumps to 7.3 nN, staying steady until the normal force reaches 220 nN, after which the friction rises sharply again. The first two jumps at normal forces of 50 nN and 120 nN match broadly with the rupture forces of layers 16 and 17 shown in Fig. 5**b**, indicating the AFM tip sliding on and then pushing through near surface lamellae. At high force, AFM cantilevers bend unpredictably, leading to non-linear compliance, as shown in the inset of Fig. 5**b**. Typically, non-linear compliance data is not shown; however, here, the inset of Fig. 5**b** shows that there could be another push-through layer (layer 18) at a normal force of ~240 nN, as it aligns well with the normal force (220 nN) required for the final jump in friction (Fig. 5**a** inset). The thicknesses of layers 16 and 17 are around 2.4 nm, which are consistent with outer $[AOT]^-$ bilayers of the same potential (~2.5 nm) and the repeat distance of bulk $[AOT]^-$ bilayers (2.6 nm),[72] strongly indicating that the AFM tip pushes through 17 $[AOT]^-$ bilayers as it approaches the stainless steel surface at +1.0 V.

The innermost detectable layer (layer 18) at +1.0 V is ~1.3 nm, 50 % thinner than the outer layers on stainless steel but consistent with the innermost layers found at positively charged Au(1 1 1) surfaces, which indicates the rupture of the upper half of the boundary hydrated $[AOT]^-$ bilayer. The lateral structure at OCP+1.0 V (Fig. 5**e**) shows reduced ripple-like structures compared to those at OCP and OCP-1.0 V with fewer prominent structures originating from the stainless steel. This is consistent with a more robust boundary layer able to resist thermal fluctuations, layer deformation and energy dissipation. Combined this leads to low friction during sliding. The mechanism of superlubricity found at the positively charged stainless steel is analogous to that observed at Au(1 1 1), which arises from AFM tip sliding over the robust upper half of hydrated $[AOT]^-$ bilayers.

Fig. 5**f** shows friction versus time plot for 1.6 M [BMIm][AOT] on stainless steel which reveals the friction response to potential change. A steady normal force of 120 nN is held while alternating surface potentials between OCP−1.0 V and OCP+1.0 V. At −1.0 V, the friction remains consistent at approximately ~200 nN as the AFM tip slides over a poorly formed $[BMIm]^+$ rich boundary layer. When the potential is switched to + 1.0 V a swift transition to superlubricity occurs (less than 15 s), which indicates that a highly ordered boundary hydrated $[AOT]^-$ bilayer forms rapidly. The alternation of high friction and superlubricity is replicable multiple times (Fig. 5**f**), showing the boundary layer is constantly refreshed. Previous molecular dynamics simulations of a Na [AOT]-water lamellar system (~40 wt% $[AOT]^-$) showed that the water channels facilitate swift transitions of $[AOT]^-$ ions from one half layer to the other half layer [101]. In our system, although the water concentration is much lower, it is still possible that water channels exist and aid the rapid transition of the boundary layer as a function of surface potential.

Recently, superlubricity has been also found for hexadecyltrimethylammonium bromide (CTAB) with a concentration of 2.7 mmol/L (three times the critical micelle concentration) on Au(1 1 1) after applying −0.8 V to the solution for 3500 s. The results from the CTAB study align well with our findings here and strongly indicate that for water-based lubricants, the formation of a well-defined hydrated surfactant bilayer is the key to switch on superlubricity [14].

4. Conclusions

Achieving and controlling superlubricity to enable energy conservation remains a substantial challenge, despite progress in the development of high-performance lubricants [8,21,22]. For the first time, superlubricity has been observed in a water-in-SAIL mixture (1.6 M [BMIm][AOT]) at positively charged Au(1 1 1) and stainless steel. It results from the AFM tip sliding over fluid water molecules in the hydration layer of well-defined $[AOT]^-$ bilayers. Superlubricity persists up to a critical normal force, beyond which friction jumps to a high level. This sudden increase in friction occurs when the AFM tip displaces the upper half of the hydrated $[AOT]^-$ bilayer; the sliding plane shifts to the lower half of the bilayer, and the AFM tip encounters the less lubricating alkyl chains of $[AOT]^-$. The magnitude of the critical normal force increases with potential, since the hydrated $[AOT]^-$ bilayer is better ordered at a more positively charged surface, thus withstanding higher normal force. Superlubricity is not observed at OCP or OCP−1.0 V because the rougher $[BMIm]^+$ enriched boundary layers are less able to templating subsequent robust $[AOT]^-$ bilayers, consistent with more pronounced ripples detected by AFM imaging and weaker push-through forces in normal force curves. Superlubricity on the stainless steel surface can be repeatedly switched on and off in real-time by simply changing the potential between OCP+1.0 V and OCP−1.0 V, indicating a swift change of boundary layer structure with potential. The outcomes of this work pave the way for the design of potential-dependent superlubricity systems for stainless steel, and dynamic friction control in smart devices.

5. Declaration of generative AI in scientific writing

During the preparation of this work the authors used Claude to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Yunxiao Zhang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hua Li:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Jianan Wang:** Writing – review & editing, Methodology. **Debbie S. Silvester:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Gregory G. Warr:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Rob Atkin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2024.08.187.

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